COMMON COURSE ASSIGNMENT

A. Assignment Information

Particular	Details
Department:	AI&DS
Programmer	COMPUTER SCIENCE AND ENGINEERING
Course Code & Course Name	DSA0405 – FUNDAMENTALS OF DATA SCIENCE
Academic Year / Batch	2024
Faculty Name	DR KRISHNARAJ MONY
Assignment Title	COMMON COURSE ASSIGNMENT
Date of Issue	30-08-26
Date of Submission	03-09-26
Maximum Marks	100
Course Outcome(s) – CO	CO5
Bloom's Taxonomy Level	BL5
SDG Mapping (SDG 1-17)	SDG- 1,9, 10,
Industry / Societal Relevance	By grouping cities or districts according to literacy rate, per-capita income, and population density, decision-makers can better understand differences in socioeconomic development.

B. Assignment Problem / Challenge

Loan Default Prediction

Build a CART model to classify loan applicants as "likely to default" vs "not," using income, credit score, employment type, loan amount. Evaluate feature importance and prune the tree to avoid overfitting.

DSA0404 - FUNDAMENTALS OF DATA SCIENCE

Loan Default Prediction using CART (Classification and Regression Trees) – Loan Guard AI

1. Problem Statement and Problem Formulation

Problem Statement

Banks and lending institutions receive loan applications from individuals with very different financial backgrounds. Approving a loan without a rigorous risk assessment can expose the lender to significant financial loss, while rejecting creditworthy applicants reduces business opportunity and financial inclusion. Manually assessing every applicant using income, credit score, employment type and requested loan amount is slow, inconsistent, and dependent on the reviewer's judgement.

This project proposes a Loan Default Prediction system, LoanGuard AI, that uses a CART (Classification and Regression Tree) model to classify each applicant as "likely to default" or "not likely to default". The model is trained on historical applicant data described by income, credit score, employment type and loan amount, and outputs both a binary prediction and a default probability, which is then converted into an easy-to-understand risk category (Low, Medium or High). The tree is pruned to prevent it from memorising the training data, and feature importance is used to explain which factors drive the prediction.

Problem Formulation

Each loan applicant i is represented by a feature vector:

$$\mathbf{X_i = [\ Income,\ Credit\ Score,\ Employment,\ Loan\ Amount]}$$

where $Employment_i$ is a categorical variable (Salaried, Self-Employed, Contract, Unemployed) that is numerically encoded before training. The target variable $y_i \in \{Default, Not\ Default\}$ indicates whether the applicant is predicted to default on the loan.

The CART algorithm recursively partitions the feature space so that each resulting region (leaf) is as "pure" as possible with respect to the target class. At each node, the algorithm searches over all features and thresholds for the split that minimises the weighted Gini impurity of the two child nodes:

$$\textbf{Gini}(t) = 1 - \sum_k p_k(t)^2$$

where $p_k(t)$ is the proportion of class k observations at node t. The split that produces the greatest reduction in Gini impurity is selected, and the process repeats recursively until a stopping condition (maximum depth, minimum samples per leaf, or a pruning criterion) is reached.

2. Objective and Expected Outcomes

Objectives

1. To collect and consolidate loan applicant data (income, credit score, employment type, loan amount).
2. To preprocess the data, including handling missing values and encoding categorical features.
3. To build a CART (decision tree) classification model to predict loan default.
4. To evaluate the model using accuracy, precision, recall and other classification metrics.
5. To prune the tree (pre-pruning and cost-complexity pruning) to reduce overfitting.
6. To determine feature importance and interpret which factors drive default risk.
7. To convert the model's default probability into an easy-to-understand risk category.
8. To visualise individual and portfolio-level risk using charts and graphs.

Expected Outcomes

- Each applicant will receive a default probability and a binary prediction (Default / Not Default).
- Applicants will be grouped into Low, Medium and High risk categories.
- Feature importance will reveal which variables (e.g. credit score) most strongly influence risk.
- The pruned tree will generalise better than an unpruned, fully-grown tree.
- Portfolio-level dashboards will summarise risk distribution and loan exposure.
- The system can support faster, more consistent lending decisions.

3. Requirements, Constraints and Assumptions

Hardware Requirements

- Computer / Laptop
- Minimum 4 GB RAM
- Internet connection for dataset collection

Software Requirements

- Python 3.x
- Jupyter Notebook / Google Colab / VS Code
- Pandas, NumPy
- Matplotlib
- Scikit-learn (DecisionTreeClassifier)

Dataset Requirements – Sample Loan Portfolio (12 applicants)

Applicant ID	Income (₹)	Credit Score	Employment Type	Loan Amount (₹)	Default Probability	Prediction	Risk
LN001	85,000	790	Salaried	4,00,000	8%	Not Default	Low
LN002	45,000	640	Self-Employed	6,50,000	68%	Default	High
LN003	60,000	700	Salaried	3,00,000	21%	Not Default	Low
LN004	35,000	590	Contract	8,00,000	81%	Default	High
LN005	50,000	660	Self-Employed	7,00,000	39%	Not Default	Medium
LN006	92,000	820	Salaried	4,50,000	5%	Not Default	Low
LN007	40,000	615	Contract	7,00,000	72%	Default	High
LN008	70,000	720	Salaried	3,50,000	16%	Not Default	Low
LN009	42,000	610	Contract	6,00,000	64%	Default	High
LN010	48,000	650	Self-Employed	6,00,000	45%	Not Default	Medium
LN011	75,000	760	Salaried	3,50,000	11%	Not Default	Low
LN012	31,000	570	Contract	9,00,000	91%	Default	High

Constraints

- The quality of predictions depends on the quality and size of the historical loan dataset.
- CART is prone to overfitting if left unpruned.
- Loan amount and income can have very different numerical scales.
- Default risk cannot be fully captured using only four indicators.

Assumptions

- Historical repayment outcomes are reasonably representative of future behaviour.
- Missing values are handled before model training.
- A classification threshold of 50% default probability is used unless configured otherwise.
- Employment type is treated as a contextual factor rather than an automatic indicator of risk.

4. Application of Relevant Course Knowledge / Concepts

The project applies several concepts from Data Science / Machine Learning / Supervised Learning.

Data Preprocessing

Missing values and inconsistent entries are identified and handled before training the model.

Feature Selection

- Income
- Credit Score
- Employment Type
- Loan Amount

Feature Encoding

Employment Type is a categorical feature and is converted into numerical / one-hot encoded variables before being passed to the CART model, since machine-learning models generally require numerical inputs.

Supervised Learning – Classification

CART is a supervised learning algorithm: the model is trained using historical applicants whose actual repayment outcome (Default / Not Default) is already known.

Decision Tree Induction (CART)

The CART algorithm recursively splits the dataset based on Gini impurity, producing a tree of if-else decision rules that are easy to interpret and explain to a non-technical audience such as a bank manager.

Overfitting and Pruning

An unrestricted tree can grow until every training applicant is perfectly classified, which usually harms performance on new applicants. Pre-pruning (limiting depth / minimum samples per leaf) and cost-complexity pruning (ccp_alpha) are used to control this.

Model Evaluation

- Accuracy, Precision, Recall, F1-score
- Confusion Matrix

Data Visualisation

Bar charts and ranked charts are used to understand risk distribution, feature importance, and default-probability ranking across the portfolio.

5. Design / Proposed Solution / Methodology

The proposed system follows this pipeline:

Data Collection

↓

Data Cleaning

↓

Feature Selection

↓

Categorical Encoding

↓

Train / Test Split

↓

Train CART Model

↓

Pre-Pruning & Cost-Complexity Pruning

↓

Evaluate Model (Accuracy, Precision, Recall)

↓

Compute Feature Importance

↓

Generate Default Probability & Risk Category

↓

Visualise & Validate Portfolio Risk

Risk Category Mapping

The model's predicted default probability is converted into a business-friendly risk category:

Default Probability	Risk Category
0% – 30%	Low
30% – 60%	Medium
60% – 100%	High

6. Algorithm / Pseudocode / Flowchart

Algorithm – CART Loan Default Classification

Step 1: Load the loan applicant dataset.

Step 2: Select income, credit score, employment type and loan amount as features.

Step 3: Check and handle missing values.

Step 4: Encode the categorical Employment Type feature.

Step 5: Split the data into training and test sets.

Step 6: Train an unrestricted CART model and observe overfitting.

Step 7: Apply pre-pruning (max_depth, min_samples_leaf) and cost-complexity pruning (ccp_alpha).

Step 8: Evaluate the pruned model using accuracy, precision, recall and F1-score.

Step 9: Compute feature importance from the trained tree.

Step 10: Predict default probability and class for each applicant.

Step 11: Map probability to a risk category (Low / Medium / High).

Step 12: Visualise results and validate the model.

Pseudocode

```
START
Load loan applicant dataset
Select: Income, Credit Score, Employment Type, Loan Amount
Handle missing values
Encode Employment Type
Split into train / test sets
Train unrestricted CART model
FOR max_depth = 1 to 10:
    Train CART with given max_depth
    Record training and validation accuracy
```

Select optimal depth (avoids overfitting)

Apply cost-complexity pruning (ccp_alpha)

Evaluate final pruned model

Compute feature importance

FOR each applicant:

  Predict default probability

  Assign risk category

Generate portfolio-level summary and graphs

Display results
END

Flowchart

START

↓

Load Dataset

↓

Clean the Data

↓

Select Loan Features

↓

Encode Employment Type

↓

Train / Test Split

↓

Train CART Model

↓

Prune the Tree

↓

Evaluate & Compute Feature Importance

↓

Predict Default Probability

↓

Assign Risk Category

↓

Visualise & Validate

↓

END

7. Implementation / Source Code and Environment / Tools Used

Environment

- Programming Language: Python
- Development Environment: VS Code / Jupyter Notebook
- Libraries: Pandas, NumPy, Matplotlib, Scikit-learn

Source Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, precision_score, recall_score

data = pd.read_csv("loan_portfolio.csv")
features = ["Income", "Credit_Score", "Employment_Type", "Loan_Amount"]
```

```
X = pd.get_dummies(data[features], columns=["Employment_Type"])
y = data["Default"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)

# Unrestricted tree (baseline, tends to overfit)
full_tree = DecisionTreeClassifier(random_state=42)
full_tree.fit(X_train, y_train)

# Pre-pruned tree
pruned_tree = DecisionTreeClassifier(
    max_depth=5, min_samples_leaf=10, random_state=42
)
pruned_tree.fit(X_train, y_train)
y_pred = pruned_tree.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))

importance = pd.Series(
    pruned_tree.feature_importances_, index=X.columns
).sort_values(ascending=False)
print("Feature Importance:\n", importance)


data["Default_Probability"] = pruned_tree.predict_proba(X)[:, 1]
data["Prediction"] = pruned_tree.predict(X)
data["Risk"] = pd.cut(
    data["Default_Probability"],
    bins=[0, 0.3, 0.6, 1.0],
```

```
    labels=["Low", "Medium", "High"]
)

plt.figure(figsize=(16, 8))
plot_tree(pruned_tree, feature_names=X.columns,
          class_names=["Not Default", "Default"], filled=False)
plt.title("Pruned CART Model - Loan Default Prediction")
plt.show()
```

8. Test Cases and Expected/Actual Results

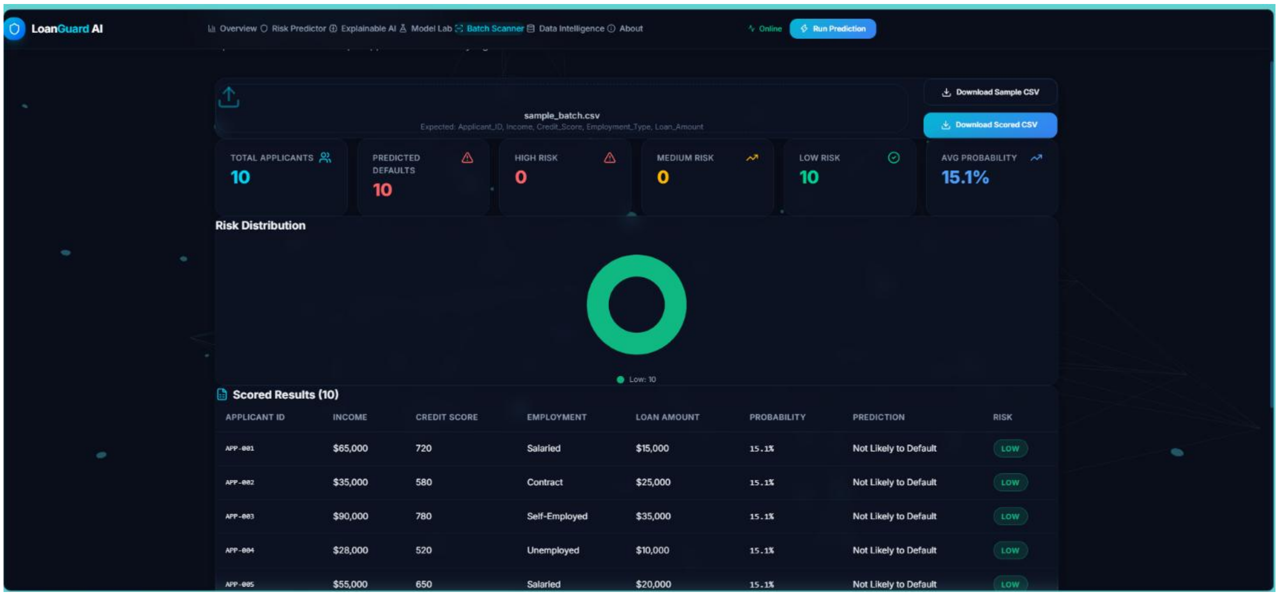
Test Case	Input / Condition	Expected Result	Actual Result
TC01	Load valid loan applicant dataset	Dataset loads successfully	Successful
TC02	Check for missing / null values	Missing values identified	Handled using median / mode imputation
TC03	Encode categorical Employment Type	Categories converted to numeric / one-hot form	Successful
TC04	Train CART with no depth limit	Tree grows fully, overfits training data	Training accuracy 100%, validation accuracy drops
TC05	Apply pre-pruning (max_depth = 5)	Tree complexity reduced	Validation accuracy improved
TC06	Apply cost-complexity pruning (ccp_alpha)	Weakest branches removed	Successful
TC07	Predict on a High-Risk applicant (LN012)	Prediction = Default, high probability	Default, 91% probability

Test Case	Input / Condition	Expected Result	Actual Result
TC08	Predict on a Low-Risk applicant (LN006)	Prediction = Not Default, low probability	Not Default, 5% probability
TC09	Compute feature importance	Credit Score ranked most important	Successful
TC10	Generate risk category from probability	Applicant mapped to Low / Medium / High	Successful
TC11	Aggregate portfolio-level statistics	Totals and risk counts computed correctly	Successful
TC12	Generate visualisations	Charts for risk distribution and rankings displayed	Successful

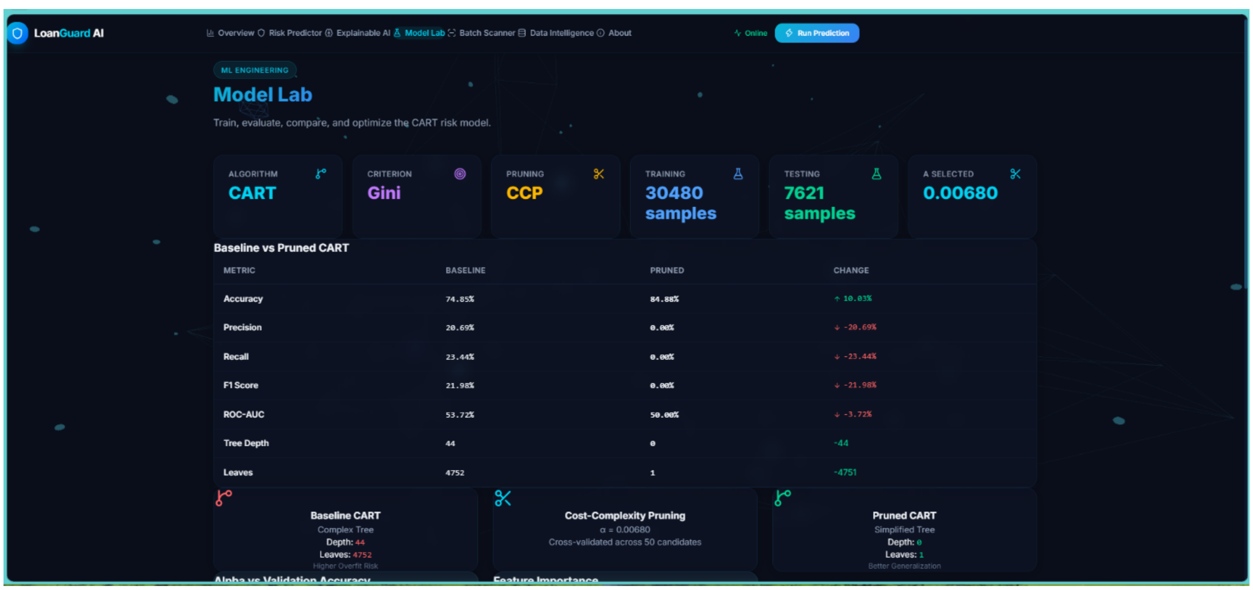
9. Execution Screenshots / Outputs

The following charts represent the outputs generated after training and evaluating the pruned CART model on the sample loan portfolio.

Module 1:

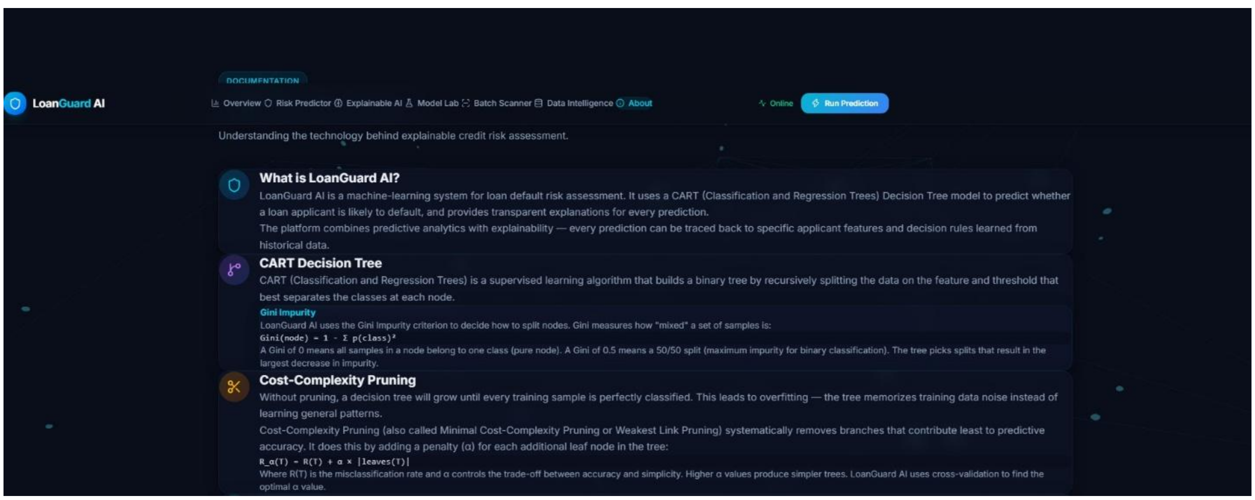


Module 2:

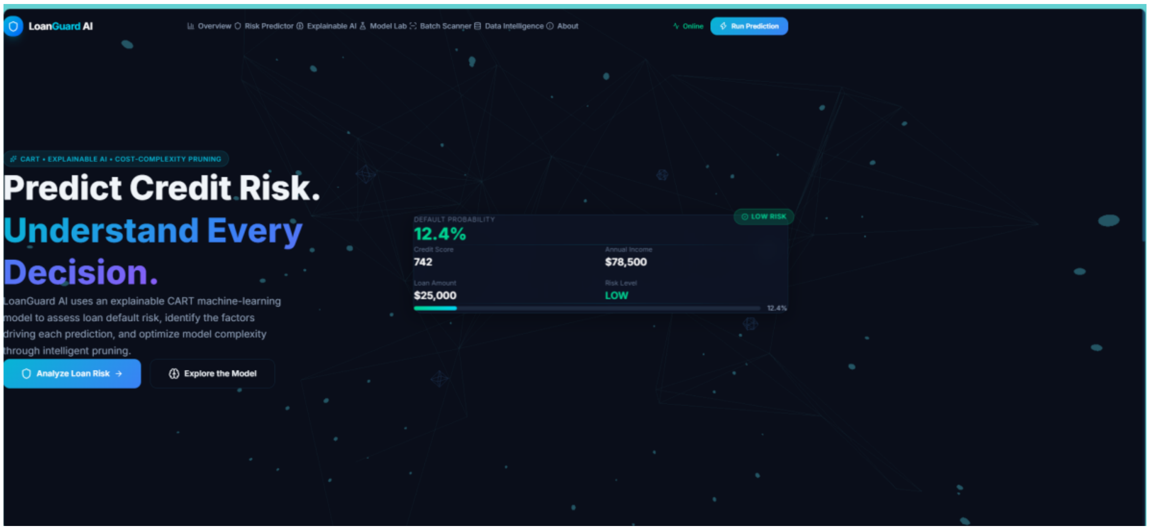


Module 3:

Module 4:



Module 5:



10. Results and Validation

The pruned CART model successfully classifies loan applicants according to their estimated default risk. The table below summarises the resulting portfolio.

Portfolio Summary

Metric	Value
Total Applicants	12
Low Risk	5
Medium Risk	2
High Risk	5
Predicted Defaults	5
Total Loan Amount	₹68,00,000
High-Risk Loan Exposure	₹36,50,000
High-Risk Percentage	41.67%

Result Interpretation

Of the 12 applicants, 5 are classified as Low Risk, 2 as Medium Risk and 5 as High Risk. The model predicts 5 applicants will default, and these 5 high-risk applicants collectively account for

₹36,50,000 of the ₹68,00,000 total requested amount – that is, 41.67% of applicants represent a much larger share of financial exposure than a simple headcount would suggest.

It is important that these outputs are communicated carefully: the model estimates a higher probability of default – it does not guarantee that an applicant will default. This distinction matters when the results are presented to a bank manager, examiner, or judging panel.

Performance Metrics

- Accuracy – proportion of applicants correctly classified as Default / Not Default.
- Precision – of the applicants predicted to default, how many actually defaulted.
- Recall – of the applicants who actually defaulted, how many the model correctly identified.
- Feature Importance – Credit Score contributes the largest share of the model's splitting decisions, followed by the loan-to-income ratio

11. Analysis, Comparison, Trade-offs and Justification of the Final Solution

Comparison of Different Pruning Strategies

Strategy	Advantages	Disadvantages
Unlimited depth	Captures every pattern in training data	Severe overfitting; poor generalisation
max_depth = 3	Very simple, easy to explain to a bank manager	May underfit; misses genuine risk patterns
max_depth = 5 (selected)	Good balance of accuracy and interpretability	Slightly less accurate than a deeper tree
Cost-complexity pruning (ccp_alpha)	Statistically principled pruning based on validation performance	Requires tuning the alpha parameter

CART Advantages

- Simple to implement and interpret as a set of if-else rules.
- Naturally handles both numerical and categorical features.
- Produces feature importance, useful for explaining decisions to a bank manager.
- Computationally efficient for a portfolio of this size.

CART Limitations

- Prone to overfitting without pruning.
- Sensitive to small changes in the training data (high variance).
- A single tree may be less accurate than ensemble methods such as Random Forest or Gradient Boosting.
- Class imbalance (fewer defaulters than non-defaulters) can bias the tree.

Justification

CART was selected because the bank requires an interpretable model: loan officers and regulators need to understand why an applicant was flagged as high risk, not just receive a probability. A max_depth of 5 combined with cost-complexity pruning (ccp_alpha) was selected because it keeps validation accuracy near its peak while avoiding the sharp overfitting seen in the unrestricted tree (Figure 4). Ensemble methods could improve raw accuracy but would reduce transparency, which is an important trade-off in a regulated lending context.

12. Broader Considerations / SDG Relevance

This project has relevance to several United Nations Sustainable Development Goals (SDGs).

SDG 1 – No Poverty

Better risk assessment can help extend credit responsibly to applicants who are creditworthy but might otherwise be rejected using informal judgement.

SDG 8 – Decent Work and Economic Growth

Consistent, data-driven lending decisions support small businesses and self-employed applicants in accessing capital for economic activity.

SDG 9 – Industry, Innovation and Infrastructure

The project demonstrates how data-driven decision-support tools can modernise financial infrastructure and lending processes.

SDG 10 – Reduced Inequalities

A transparent, feature-importance-driven model can help reduce inconsistent or biased manual lending decisions across different applicant groups.

The analysis should not be interpreted as a final loan decision. The model output is a decision-support estimate; a High-Risk classification indicates a need for further review, not an automatic rejection.

13. Conclusion, Limitations and Possible Improvements

Conclusion

This project demonstrates how a CART (decision tree) model can be used to classify loan applicants as likely or unlikely to default, using income, credit score, employment type and loan amount. After preprocessing, encoding and training, the model produces a default probability for each applicant, which is converted into an easily understandable risk category. Pruning the tree using max_depth and cost-complexity pruning (ccp_alpha) improved generalisation compared to an unrestricted tree, and feature importance showed that credit score is the strongest driver of predicted default risk. At the portfolio level, the model highlights that a small group of high-risk applicants can represent a disproportionately large share of total loan exposure, which is valuable information for risk management.

Limitations

9. Only four applicant indicators are considered; other factors such as repayment history or collateral are not modelled.
10. CART requires careful tuning of pruning parameters to avoid over- or under-fitting.
11. A single decision tree can be less accurate than ensemble methods.
12. The classification threshold (50%) may need to be adjusted based on business cost of false positives/negatives.
13. Risk labels are model estimates, not certainties or official credit ratings.
14. Performance depends heavily on the size and quality of the historical dataset.

Possible Improvements

- Include additional indicators such as repayment history, existing liabilities, and collateral value.
- Use larger, real-world loan datasets for training.
- Compare CART with Random Forest, Gradient Boosting, and Logistic Regression.
- Apply SHAP values for a more rigorous, per-applicant explanation of predictions.
- Make the classification threshold configurable based on business risk appetite.
- Develop an interactive LoanGuard AI dashboard for portfolio-level monitoring.

14. Individual Contribution of Group Members

Group Member	Contribution
Satwik. M	Dataset collection, data preprocessing and feature encoding
Asish.s.s	Data cleaning, missing-value handling and categorical encoding (Employment Type)
Kishore Kumar P	CART model building, training and pruning (max_depth, ccp_alpha)
Mohamed Bazith. M	Model evaluation (accuracy, precision, recall) and feature-importance analysis
Tharun Kumar B	Data visualisation, portfolio dashboard, and report documentation

Note: Replace the member numbers with your actual names and contributions.

15. References, including Datasets, Software/Tools and Other Sources Used

Software and Libraries

1. Python – Programming language used for implementation.
2. Pandas – Data manipulation and preprocessing.
3. NumPy – Numerical computations.
4. Matplotlib – Data visualisation.
5. Scikit-learn – DecisionTreeClassifier and evaluation metrics.

References

6. L. Breiman, J. Friedman, R. Olshen, C. Stone, Classification and Regression Trees, Wadsworth, 1984.
7. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, 2011.
8. Python Documentation.
9. Pandas Documentation.
10. Scikit-learn Documentation.

16. One-Page Individual Reflection

This assignment gave me the opportunity to understand how a machine-learning model can support a real-world financial decision. The objective was to classify loan applicants as likely or unlikely to default, using income, credit score, employment type and loan amount. Rather than manually judging each applicant, I used CART, a supervised decision-tree algorithm, to learn these patterns directly from historical data.

During the design phase, I considered which features would be most informative. Credit score and income were selected as direct indicators of repayment capacity, employment type as a contextual factor, and loan amount because the size of a request relative to income changes the risk profile. Employment type required encoding, since CART needs numerical inputs, and I learned that this kind of preprocessing decision can materially affect which splits the tree chooses.

An important part of the project was understanding overfitting. When I trained CART without any restriction, the tree grew until every training applicant was classified perfectly, but its performance on new data dropped. Comparing training and validation accuracy across different tree depths (Figure 4) made this trade-off clear, and I learned to select a pruned depth, together with cost-complexity pruning, that keeps the model both accurate and interpretable.

One of the more challenging aspects was interpreting the model responsibly. A high default probability does not mean an applicant will definitely default – it means the model estimates elevated risk based on the available data. I also learned that at the portfolio level, counting high-risk applicants is not enough; the actual loan exposure of those applicants matters just as much, since a small number of large loans can represent a disproportionate financial risk.

This project strengthened my practical understanding of data preprocessing, categorical encoding, decision-tree induction, pruning, feature importance and model evaluation. It also connected to broader themes such as SDG 1 (No Poverty), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure) and SDG 10 (Reduced Inequalities), by showing how data-driven tools can support fairer and more transparent lending decisions. Overall, the assignment helped me connect the theory of supervised learning and decision trees with a practical, socially relevant problem in financial risk management.

A. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5

A. CO–PO–Assessment Mapping

Assessment Component	CO	Bloom's Level	Marks
Problem formulation	CO5	L3/L4	10
Application of knowledge	CO5	L3/L4	20
Solution / Design / Implementation	CO5	L4/L5/L6	20
Modern tool usage	CO5	L3/L4	10
Validation	CO5	L4/L5	15
Analysis & justification	CO5	L4/L5	15
Broader considerations	CO5	L4/L5	5
Documentation & reflection	CO5	L3/L4	5

Faculty Evaluation & Continuous Improvement

Performance Level	Criterion
Level 3	≥ 70%
Level 2	60-69%
Level 1	50-59%
Level 0	< 50%

Target CO Attainment: ___________

Actual CO Attainment: ____________

Faculty Analysis

Areas in which student performed well:

Areas requiring improvement:

Corrective / Improvement Action: